

Presentation: Robust End-to-End Image Transmission with Residual Learning

Time Allotment: 12 Minutes

1. Title Slide (0.5 minutes)

- Good afternoon, everyone. My name is Cenk Yetis, and I'm excited to present our work on 'Robust End-to-End Image Transmission with Residual Learning'.
- This research was conducted and finalized during my affiliation at the Huawei Technologies Research Center in Lund, Sweden.

2. Introduction & Motivation (2 minutes)

Let's start by understanding why this work is critical.

- Shift to Deep Learning in Image Transmission: Deep learning (DL)-based image transmission at the physical layer (PL) has shown significant outperformance compared to conventional digital transmissions.
- Challenges with PL Solutions: Implementing PL solutions requires major shifts in established standards, like those in cellular communications.
- Application Layer (AL) as a Solution: Application layer (AL) solutions offer a more feasible and standards-compliant alternative.
- 6G and Semantic Communication (SC): 6G aims for extreme requirements, and SC is a crucial step towards human-like intelligence, drastically reducing data transmission by focusing on meaning rather than bits, improving efficiency and performance.
- Problem with Existing Residual Coding NN Solutions: Existing neural network (NN) solutions with residual coding often use conventional compression techniques (like BPG and JPEG) or transmit residuals without specific channel considerations.
- Our Solution: Deep Semantics Source Channel Coding (DS2C2-DeeEsSquareCSquare): We propose a layered image transmission scheme at the AL that is robust to end-to-end (E2E) channel errors. DS2C2 is an E2E NN solution

for layered image transmission with residual coding, specifically designed for robustness in error-prone channels.

3. Key Contributions (1.5 minutes)

Now, let's have a look at the key contribution of DS2C2.

- **Proposed DS2C2:** We propose a novel deep learning-based semantic-empowered multi-layer image transmission scheme operating entirely at the application layer, ensuring a standards-compliant approach for next-generation networks.
 - **Robustness through Latent Representation:** Our solution achieves high robustness to E2E channel errors by strategically mapping the residual image into a latent representation whose structure aligns well with channel characteristics.
 - **Disjoint Architecture:** The solution introduces a disjoint architecture comprising Recursive Generative Adversarial Network (RGAN) and Block Residual Network (BResNet) (+SumNet at the receiver), providing flexibility and adaptability.
 - **Demonstrated Effectiveness:** Numerical and visual results demonstrate the effectiveness and robustness of DS2C2 under various channel error conditions and its capability for efficient updating with a low number of training images.
 - **Key Differentiators from Previous Works:**
 - DS2C2 is a complete NN solution where NN is used for encoding the residual image (BResNet).
 - Implements two consecutive Generative Adversarial Networks (GANs), referred to as RGAN.
 - Uses NN (SumNet) to improve summation accuracy at the receiver.
 - Employs a disjoint design of RGAN and BResNet (+ SumNet), allowing separate training and modularity.
 - Assumes the base layer image is received without error, focusing protection on residual information.
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4. DS2C2 Architecture and System Model (3 minutes)

- Overall Architecture (Refer to Fig. 1a):
 - Base Layer: Base layer produces s and c , r is the enhancement layer.
 - SegNet generates semantic images (s), and CompNet generates a downsampled image (c) which is upsampled to c' .
 - s and c' are fed into FineNet, whose output (f) is summed with c' to get the synthesized image x' .
 - The residual image (r) is obtained by subtracting x' from the original image and then encoded by BResNet.
 - The receiver implements operations complementary to those at the transmitter.
 - At the receiver, SumNet is included to improve summation accuracy.
 - Quantization and interleaving are applied after BResNet (details in Fig. 2).
- RGAN Architecture (Refer to Fig. 1b):
 - Comprises FineNet (generator) and DiscNet (discriminator) during training. DiscNet is not used during testing.
 - Increasing RGAN steps improves synthesized image quality at higher computational cost.
- Quantization and Interleaver (Refer to Fig. 2 and also Page 7):
 - Quantization and entropy coding improve rate-distortion performance.
 - Uses statistical binarizer for fixed bits per pixel (BPP) analysis.
 - Assumes Block Binary Erasure Channel (BBEC) model for channel errors, where entire blocks can be erased.
 - A pixel-level interleaver is used to mitigate block losses.
- BResNet Architecture (Refer to Fig. x) (Not in the presentation):
 - Composed of ResNet sub-blocks, which are 2D convolutional NNs with residual connections.

- o Uses Generalized Divisive Normalization (GDN) and Inverse GDN (IGDN) for layer normalization.
 - o Employs pixel shuffle for upsampling instead of transpose convolution.
 - o SumNet uses 3 consecutive ResNet sub-blocks.
 - Loss Function (Not in the presentation): For RGAN, $LRGAN = LG + LD$, where Ld (distance measure) includes L1-norm, SSIM, and VGG loss. The Ld metric is also used for BResNet and SumNet.
 - Training Process (Not in the presentation): RGAN is trained first by minimizing LRGAN, then the pre-trained RGAN is loaded for joint training of BResNet and SumNet by minimizing LBResNet+LSumNet.
 - Robustness Mechanism (Refer to Fig. 3):
 - o This slide visually explains why our DS2C2 algorithm exhibits high robustness against channel errors.
 - o On the left and right sides, we see r_{enc} before and after the channel, respectively
 - o The residual image is mapped into a latent representation that aligns well with E2E channel characteristics, achieving robustness against channel errors.
 - o Figure 3 illustrates this by showing similar noisy structures in the latent representation before and after channel errors.
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5. Numerical Results (3 minutes)

- Here, we present the experiments conducted using Cityscapes, ADE20K, and Kodak datasets.
- This slide numerically benchmarks our solution vs. Learned image compression with Gaussian Mixture Models and Attention Modules abbreviated by LIC-GMM-AM.
- Petrain = 0 or 8 means that the probability of error during training phase is set to 0% or 8%, meaning 0 or 8% of the bits are corrupted by the channel.

- petrain is set to 0% mean, during the training phase, channel errors are not introduced. In other words, NNs are optimized solely for image reconstruction, without any explicit training for the channel error correction.
- Benchmarking with LIC-GMM-AM (Refer to Table I):
 - LIC-GMM-AM (Learned image compression with Gaussian Mixture Models and Attention Modules) lacks channel error correction unless trained against them.
 - Without channel errors introduced during training (petrain=0, petest=16), LIC-GMM-AM's PSNR drops significantly (to -1.18 from 30.61).
 - Even with channel errors introduced during training (petrain=8, petest=16), LIC-GMM-AM only achieves 24.14 PSNR.
 - Whereas, DS2C2 achieves 25.03 PSNR for (petrain=0, petest=16) and 27.32 PSNR for (petrain=8, petest=16).
 - Key finding: DS2C2 significantly reduces image quality degradation from around 20% to just 3% compared to a key competitive NN solution.
- Low Number of Training Images:

This slide demonstrates our solution's training efficiency, showcasing its ability to reconstruct images effectively with a low number of training examples

- Figure 4 visually demonstrates effective updates even with a low number of training images.
- On the left and right sides, you see estimated images at the receiver with NNs being trained by only 16 and 2968 images, respectively.
- As you see, with only 16 training images, sufficient details are recovered for many tasks, such as object detection (car, passengers)
- Therefore, DS2C2 can be quickly updated with a small number of training images.
- This is crucial for swift updates due to changing conditions or insufficient initial training, and for situations where collecting new training data is challenging.

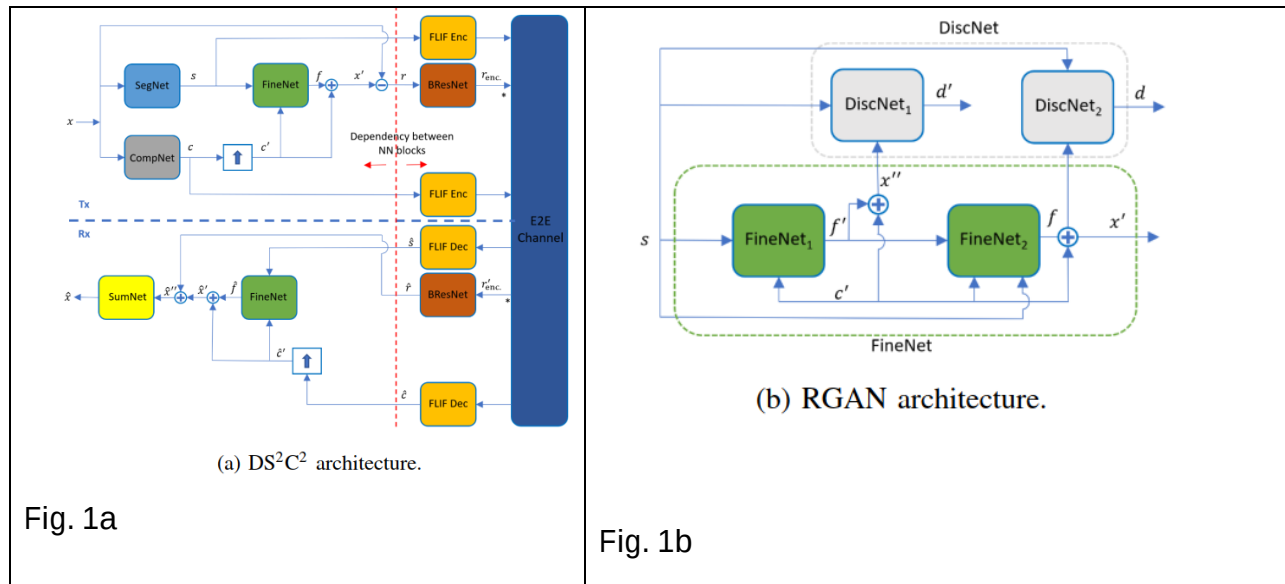
6. Conclusion & Future Work (1 minute)

- Summary: Presented DS2C2, a robust deep learning-based semantic-empowered multi-layer image transmission scheme, resilient to E2E channel errors and viable for AL implementation with moderate standard changes.
- Flexibility and Adaptability: The disjoint design offers adaptability across various scenarios under channel errors.
- Future Work: Potential expansions include extending DS2C2 to video transmission, exploring diverse and realistic channel models, and optimizing the system for specific hardware constraints to enhance deployment feasibility.

7. Student Project/Thesis Opportunity (0.5 minute)

- A Note to Graduate Students: Interested in semantic communications for your project or thesis? Please feel free to connect with Prof. Hasan Fehmi Ateş, who is attending the conference in person.

FIGURES AND TABLES



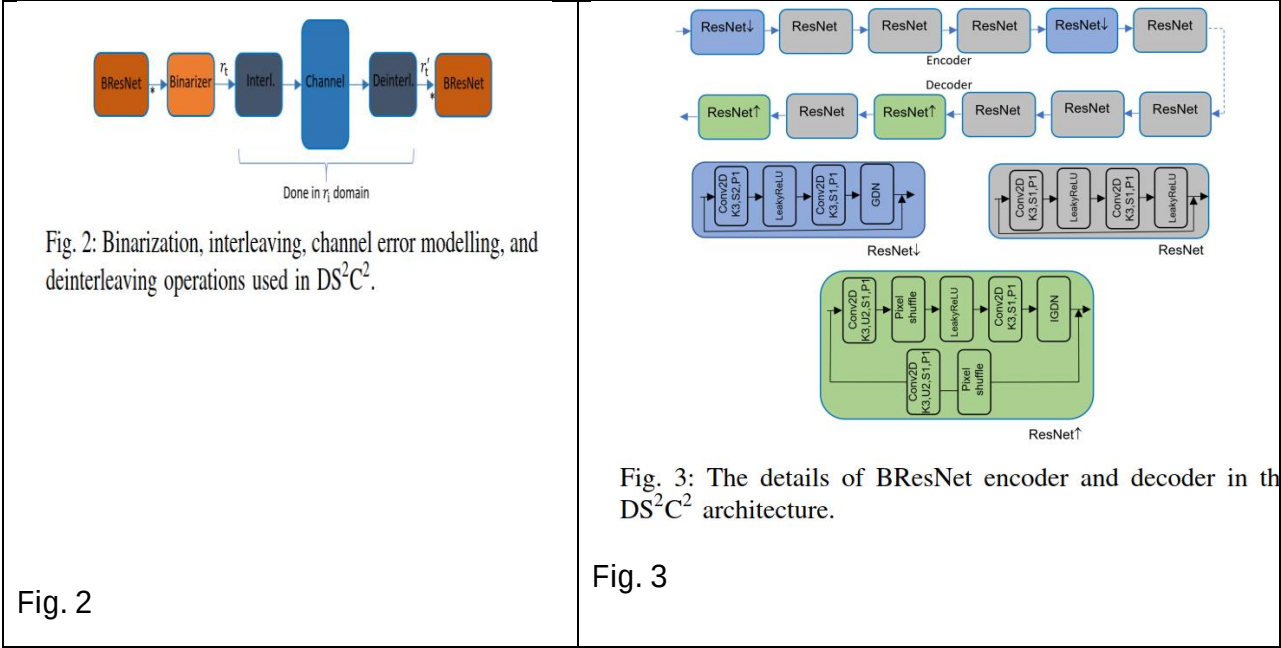


Fig. 4

TABLE I: DS^2C^2 vs. LIC-GMM-AM for Kodak dataset at 0.26 BPP. Benchmark case: $p_{e_{train}} = 0$, $p_{e_{test}} = 0$. Trained without channel errors: $p_{e_{train}} = 0$, $p_{e_{test}} = 16$. Trained with channel errors: $p_{e_{train}} = 8$, $p_{e_{test}} = 16$.

PSNR (Loss %)	Benchmark	Trained without channel errors	Trained with channel errors
DS^2C^2	28.25	25.03 (11.4%)	27.32 (3.3%)
LIC-GMM-AM	30.61	-1.18 (104%)	24.14 (21.14%)

Table I